中国人民大学 RENMIN UNIVERSITY OF CHINA

中国人民大学信息学院 SCHOOL OF INFORMATION RENMIN UNIVERSITY OF CHINA

DACon

自进化数据智能体：迈向 Data-Ontology-Agent 协同进化

张绍磊

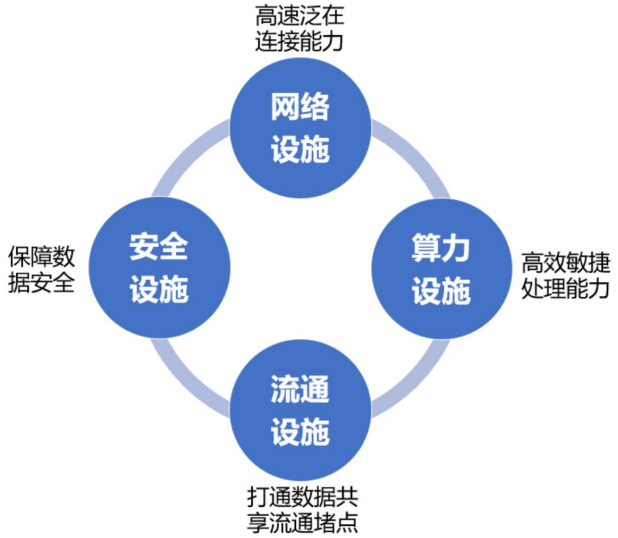
中国人民大学

2026.10.24

数据系统是释放数据价值的关键

数据系统 提升数据科学系统的智能化水平与性能，高效释放数据价值

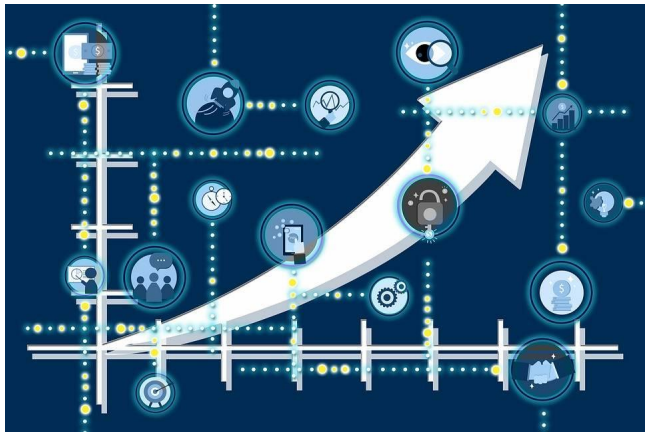
助力数据基础设施建设



促进智慧城市精细治理



推动产业数智化转型



数据科学技术起到了重要的基础性支撑作用

大数据 — 大数据 — 大数据

来源：《国家数据局数据技术设施体系》《数据空间发展战略蓝皮书》

传统数据系统

Human-first 人类专家根据领域知识处理海量数据



AI赋能的数据系统

引入Agent Agent替代人类专家，根据领域知识处理海量数据

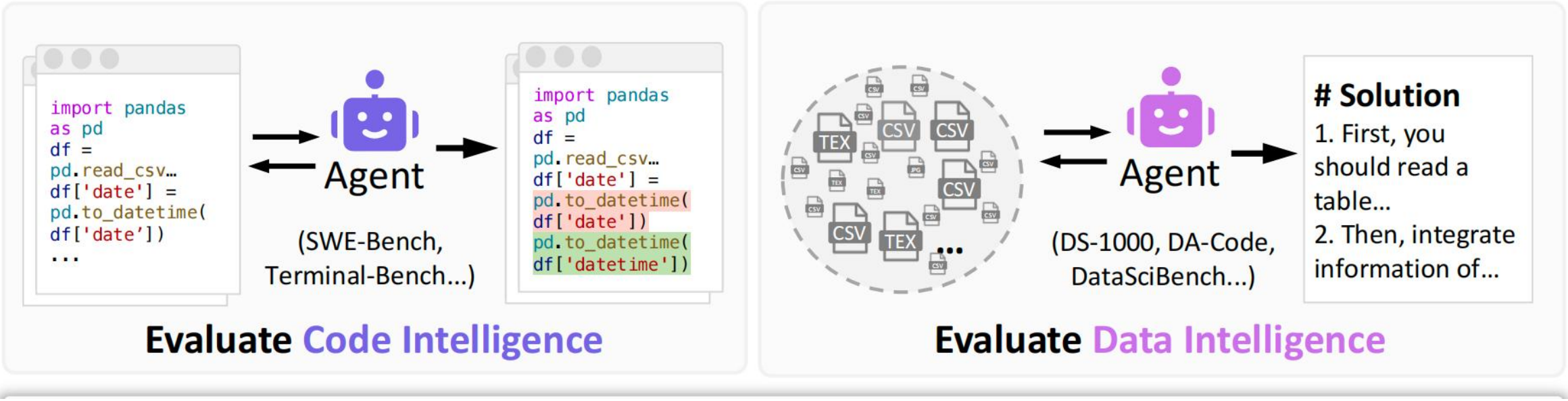
Codex、Claude Code、
Mini-swe-agent



Agent 能在海量数据中完成复杂任务么？

CoDA-Bench：Agent迷失在复杂数据环境

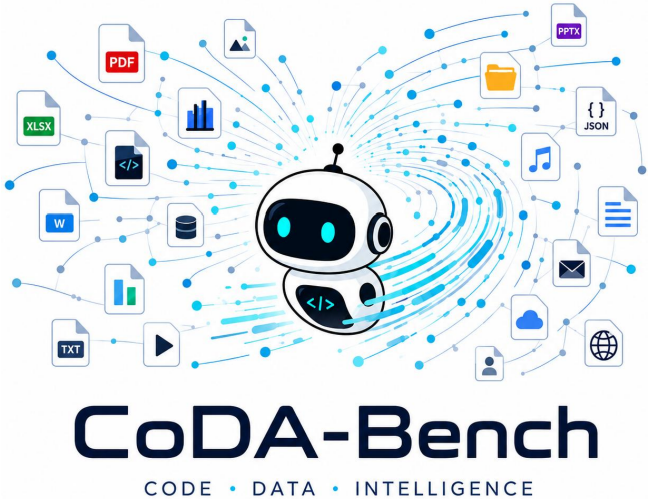
CoDA-Bench 评估Agent能不能在复杂数据环境中完成任务？



Yuxin Zhang, Ju Fan, Meihao Fan, Shaolei Zhang*, Xiaoyong Du. CoDA-Bench: Can Code Agents Handle Data-Intensive Tasks?. ICML 2026.

CoDA-Bench：Agent迷失在复杂数据环境

CoDA-Bench 评估Agent能不能在复杂数据环境中完成任务？



Yuxin Zhang, Ju Fan, Meihao Fan, Shaolei Zhang*, Xiaoyong Du. CoDA-Bench: Can Code Agents Handle Data-Intensive Tasks?. ICML 2026.

CoDA-Bench：把Agent放进海量数据环境

从 Kaggle 社区和 Notebook 中构建**数据共现图，构建海量数据社区**



数据社区

1,009	119	31	980
Tasks	**CODA-HARD**	**Communities**	**Avg Files**
full benchmark	hard subset	Kaggle-derived workspaces	per environment

EXAMPLE INSTANCE 1 / 3

Among countries with Medal_points > 5 (Medal_points = Gold·3 + Silver·2 + Bronze·1 per event) in any single Summer Olympic Games after 1993, which country had the highest rate of unique participants per 100,000 people in that specific year? (Use the country's population for that year.)

Community	community_27
Notebook	olympic-medals-vs-population
Expected answer	New Zealand; 4.18

ANSWER FORMAT

Format: Country Name; Rate. The rate is a number rounded to two decimal places.

EVIDENCE FILES (2)

```
120_years_of_olympic_history_athletes_and_results/source/athlete_events.csv
world-bank-data-1960-to-2016/source/country_population.csv
```

← Previous　Next →

SANDBOX FILE SYSTEM · COMMUNITY_27

The agent searches a real workspace — no oracle paths.

119	8,533	13.0 GB
DATASETS	FILES	ON DISK

.csv 6,195　.pdf 1,950　.json 129　.png 46　.dbf 10

```
full_community/                                  119 datasets · 8,533 files
  120_years_of_olympic_history_athletes_and_re…  3 files
    source/
      athlete_events.csv
      noc_regions.csv
      dataset-metadata.json
  countries-population/                          2 files
  population/                                    1 file
  population-by-country-2020/                    2 files
  world_population/                              7 files
  world-population-19602018/                     7 files
  world_population_by_countries/                 4 files
  gdp-world-bank-data/                           17 files — has Population…
  health-nutrition-and-population-statistics/    2 files
… … 109 more datasets
  world-bank-data-1960-to-2016/                  4 files
    source/
      country_population.csv
      fertility_rate.csv
      life_expectancy.csv
      dataset-metadata.json
```

evidence (needle)　semantically similar decoy

CoDA-Bench：Agent迷失在复杂数据环境

https://coda-bench.github.io/

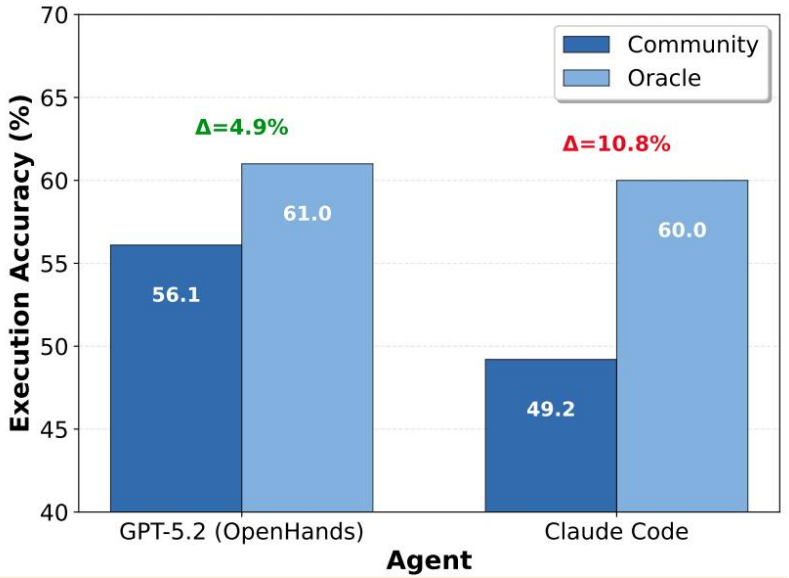
Leaderboard

Execution Accuracy (EA) is the primary score. Discovery Accuracy (DA) is the supporting signal for whether the agent found the right evidence.

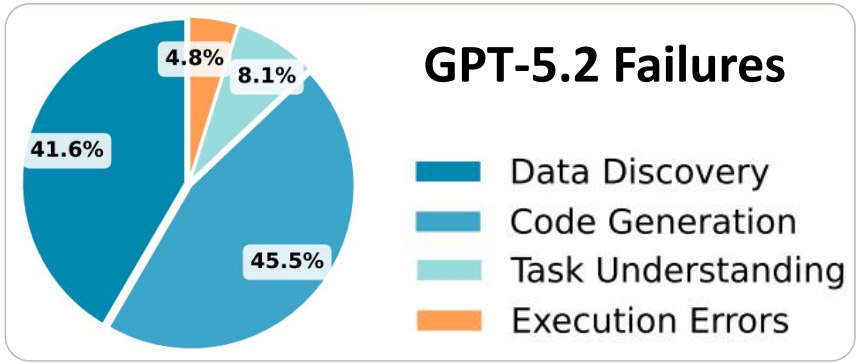
CODA-BENCH | CODA-HARD

Rank	Agent framework	Backbone	DA	EA	Tokens / task	Cost / task
1	Mini-SWE-agent	GPT-5.5	67.50%	**49.60%**	101,017	$0.39
2	Codex CLI	GPT-5.5	61.40%	**47.90%**	380,558	$1.39
3	Claude Code	Claude-Opus-4.6	68.40%	**45.40%**	113,433	$0.20
4	Claude Code	Claude-Opus-4.7	61.40%	**45.40%**	123,699	$0.22
5	OpenHands	GPT-5.5	63.20%	**44.50%**	154,913	$0.65
6	Claude Code	Claude-Sonnet-4.6	61.40%	**42.90%**	81,714	$0.11
7						42
8						7
9						41
10	OpenHands	DeepSeek-V4-Pro	58.80%	**36.10%**	330,161	$0.15

Agent 难以理解环境中海量数据间的关系



Even with the correct data, the accuracy only about 60.5%.



AI赋能的数据系统

当前局限性 数据环境复杂导致SOTA Agent系统失败



新一代数据系统：Human-first到Agent-first

Agentic Data System Agent + Ontology + Environment



自主性：完成长程任务

自进化：系统持续提升

Agent - Ontology – Data 协同进化

DeepAnalyze SkillAdam EvoOntology DeepPrep



自主性：完成长程任务

自进化：系统持续提升

DeepAnalyze：自主数据科学大模型

DeepAnalyze 模拟数据科学家的行为，自动从数据源中提取insight

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

Shaolei Zhang [1] Ju Fan [1] Meihao Fan [1] Guoliang Li [2] Xiaoyong Du [1]

ruc-datalab/DeepAnalyze 🤗 DeepAnalyze-8B 📚 DataScience-Instruct-500K 🌐 ruc-deepanalyze.github.io

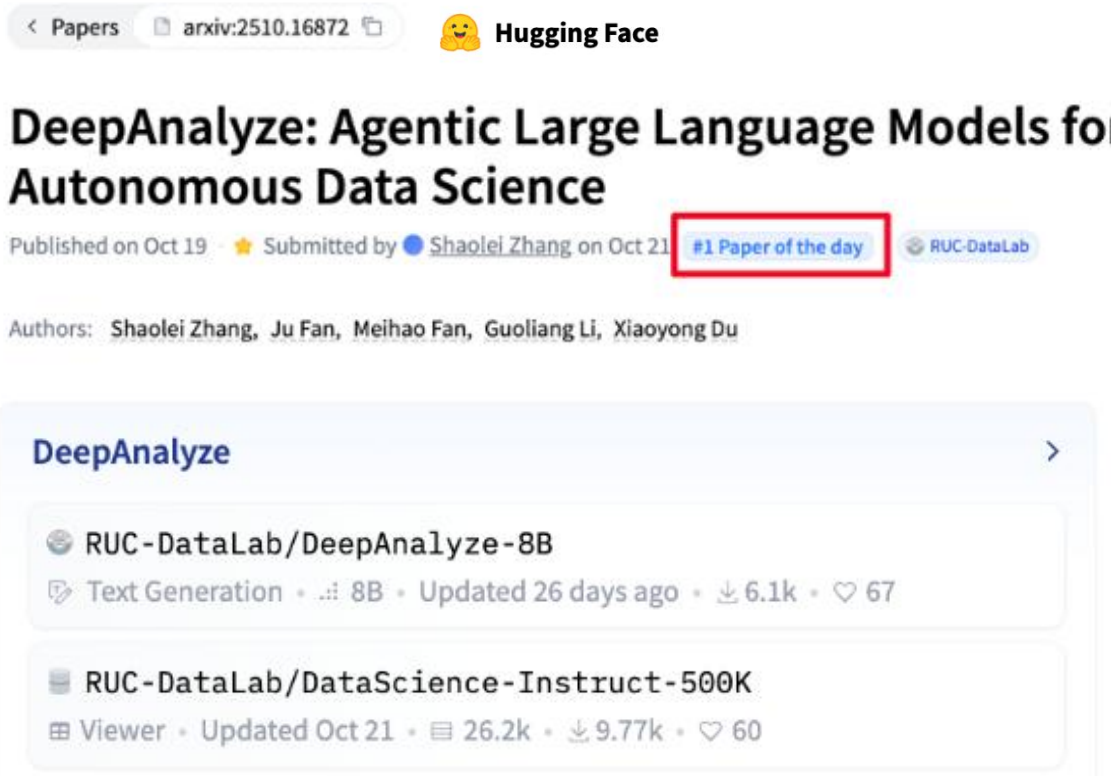
给定数据源，DeepAnalyze自主编排操作、根据反馈自适应优化

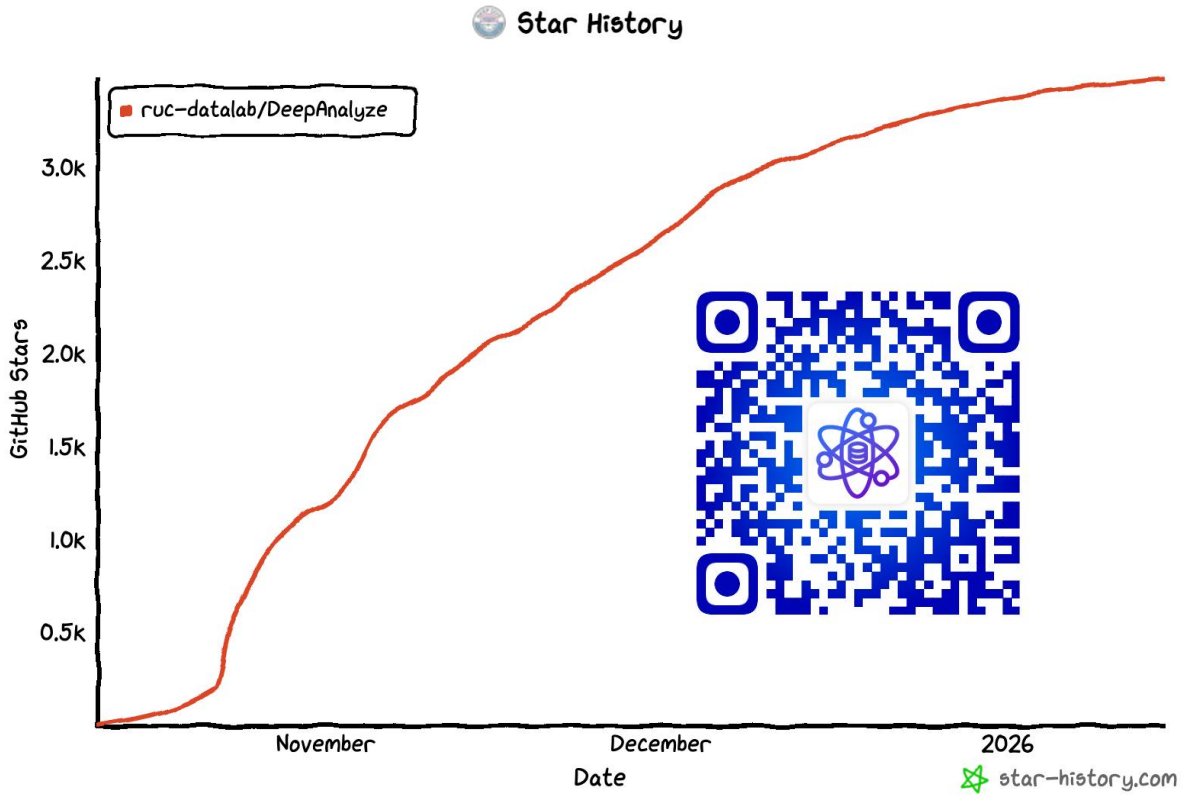
- **全流程数据科学Pipeline**：数据准备、分析、建模、可视化、洞察
- **开放式数据研究**：自动研究数据，生成分析师级别报告

DeepAnalyze：自主数据科学大模型

在HuggingFace上开源模型下载**30K+**，开源数据集下载**20K+**，曾在Huggingface日趋势榜**论文排名第一**

DeepAnalyze代码于2025年10月21日在GitHub上正式开源，已获得**4700K+**个GitHub星标





DeepAnalyze引起社区广泛讨论

Brian Roemmele @BrianRoemmele · 13h
BOOM!

Powerful scientific research now performed by your local AI is here!

Garage researchers with no pedigree can now lift humanity to new height of discovery and we ain't asking permission.

I have been testing **DeepAnalyze**-8B, an open-source agentic language model and am
Show more

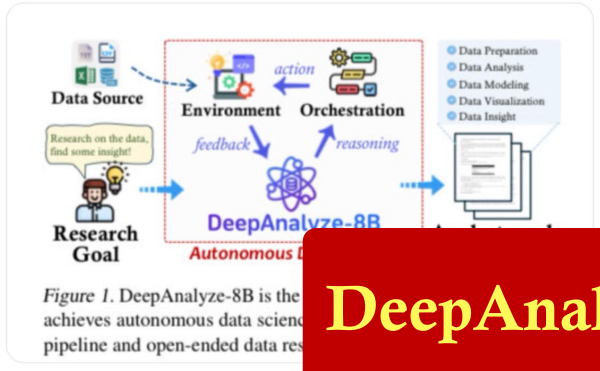
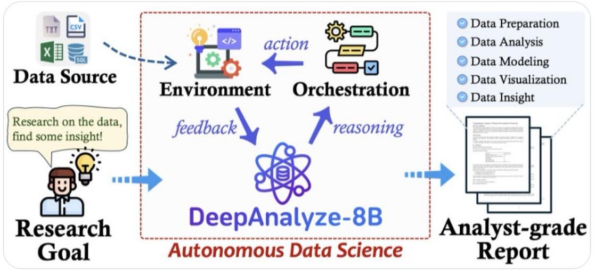


Figure 1. DeepAnalyze-8B is the
achieves autonomous data scienc
pipeline and open-ended data res

25 74

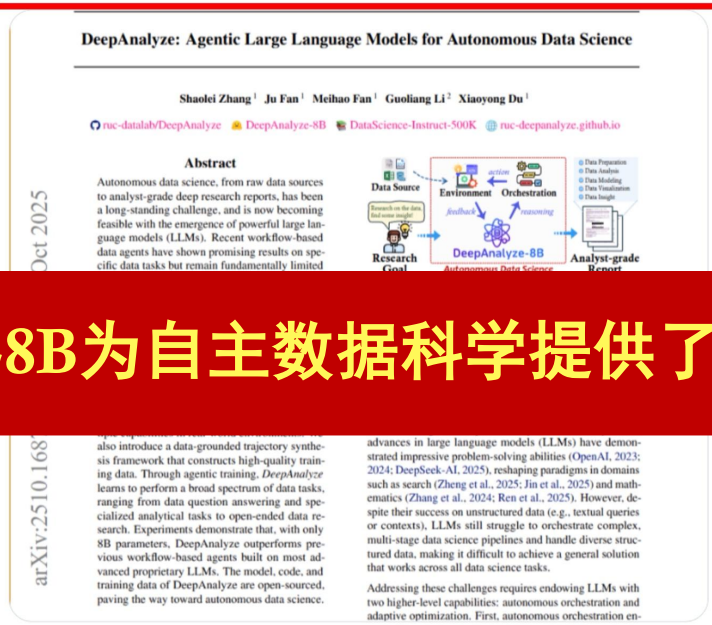
Gorden Sun @Gorden_Sun · Oct 21
DeepAnalyze：数据分析智能体
能自主完成一系列数据任务，包括：准备、分析、建模、可视化、生成报告。
Github：github.com/ruc-datalab/De...



3 69 219 16K

Dr Singularity @Dr_Singularity · 14h
Science is rapidly getting automated, and that means a 1000x acceleration is near.

DeepAnalyze-8B, the first agentic LLM designed for autonomous data science, capable of automatically completing the end to end pipeline from data sources to analyst grade deep research report.


DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

Shaolei Zhang[1] Ju Fan[1] Meihao Fan[1] Guoliang Li[2] Xiaoyong Du[1]

ruc-datalab/DeepAnalyze DeepAnalyze-8B DataScience-Instruct-500K ruc-deepanalyze.github.io

Abstract

Autonomous data science, from raw data sources to analyst-grade deep research reports, has been a long-standing challenge, and is now becoming feasible with the emergence of powerful large language models (LLMs). Recent workflow-based data agents have shown promising results on specific data tasks but remain fundamentally limited

arXiv:2510.16872v1 [cs.AI] 19 Oct 2025

also introduce a data-grounded trajectory synthesis framework that constructs high-quality training data. Through agentic training, *DeepAnalyze* learns to perform a broad spectrum of data tasks, ranging from data question answering and specialized analytical tasks to open-ended data research. Experiments demonstrate that, with only 8B parameters, DeepAnalyze outperforms previous workflow-based agents built on most advanced proprietary LLMs. The model, code, and training data of DeepAnalyze are open-sourced, paving the way toward autonomous data science.

advances in large language models (LLMs) have demonstrated impressive problem-solving abilities (OpenAI, 2023; 2024; DeepSeek-AI, 2025), reshaping paradigms in domains such as search (Zheng et al., 2025; Jin et al., 2025) and mathematics (Zhang et al., 2024; Ren et al., 2025). However, despite their success on unstructured data (e.g., textual queries or contexts), LLMs still struggle to orchestrate complex, multi-stage data science pipelines and handle diverse structured data, making it difficult to achieve a general solution that works across all data science tasks.

Addressing these challenges requires endowing LLMs with two higher-level capabilities: autonomous orchestration and adaptive optimization. First, autonomous orchestration en-

Rohan Paul @rohanpaul_ai · 23h



This paper builds an agentic LLM that can run the whole data science workflow by itself.

It is an 8B model that plans work, reads structured files, writes and runs code, checks results, and iterate...

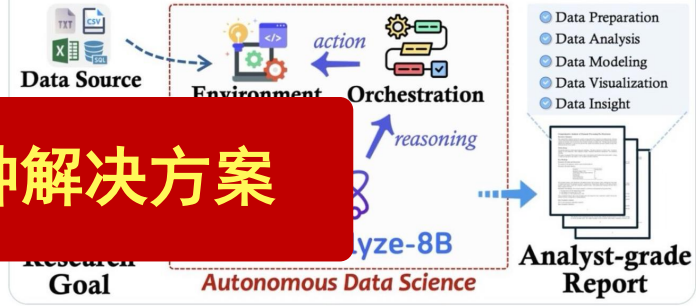
26 77 480 30K

Shaolei Zhang
@Vily_1998

Welcome everyone to use and submit issues & pull requests — we're very happy to collaborate with you all in contributing to DeepAnalyze.

Python Developer @Python_Dv · 16h
DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

DeepAnalyze is the first agentic LLM for autonomous data science. It can autonomously complete a wide range of data-centric tasks without human intervention, supporting:...
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11:41 AM · Oct 22, 2025 · **28** Views

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DeepAnalyze-8B为自主数据科学提供了一种解决方案

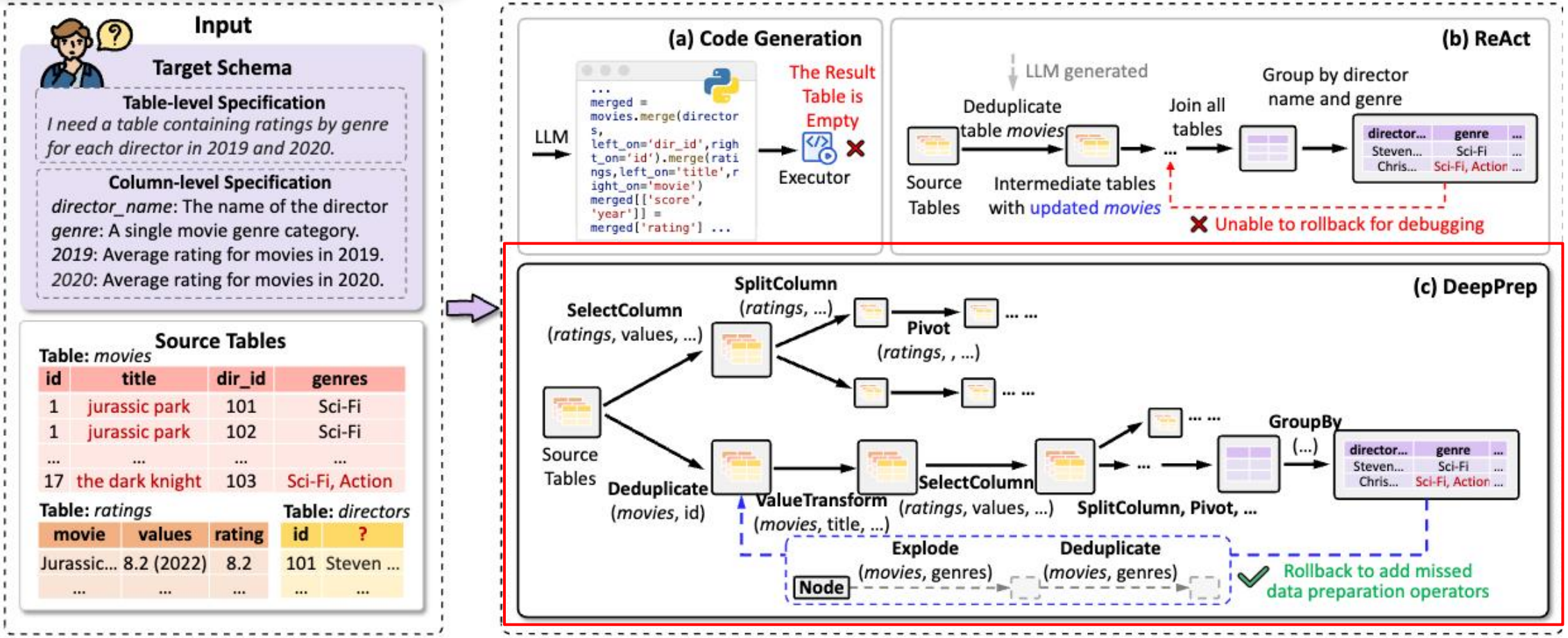
社区反馈
DeepAnalyze在数据分析上显著优于通用模型上构建的智能体

光影织梦 @SunMousekomre · 20m
We tested it. The fine-tuned model has an accuracy rate that is 20-30% higher than the agent analysis we did based on the general model.

1 4

DeepPrep：自主数据准备

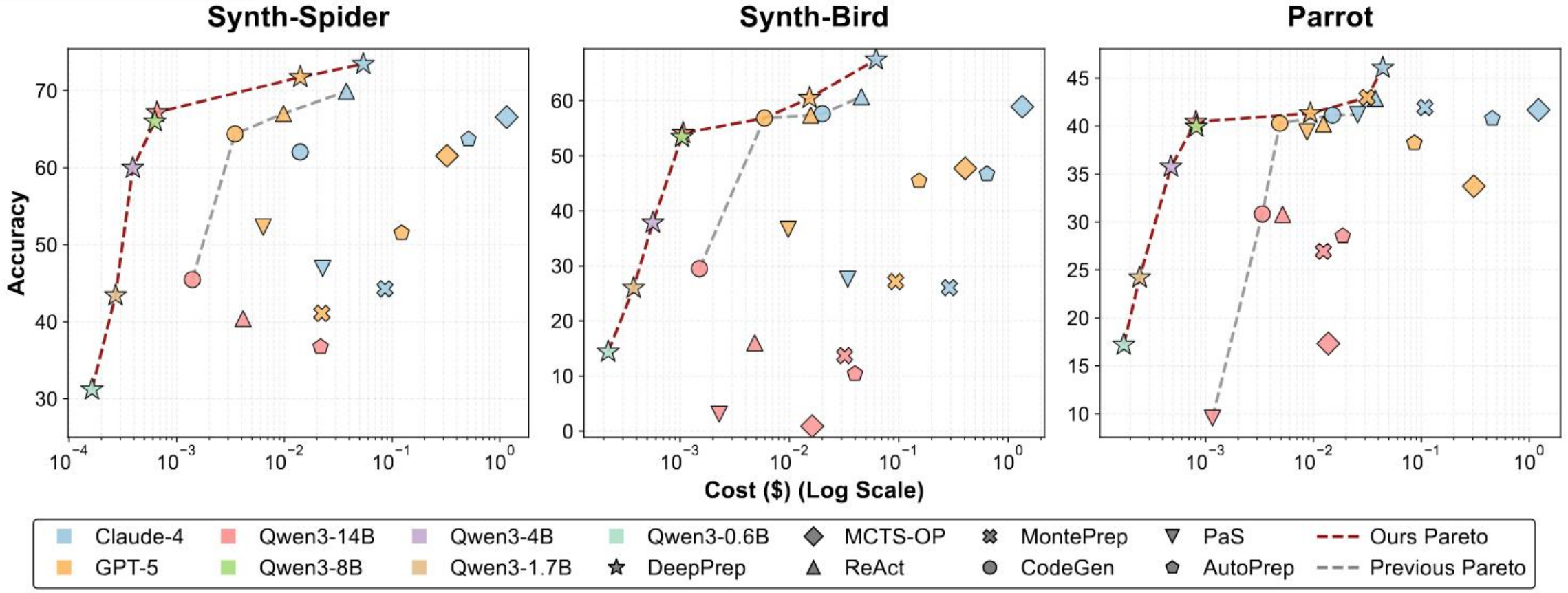
基于算子的数据准备　自然语言控制表格数据处理、转换



Meihao Fan, Ju Fan, Yuxin Zhang, Shaolei Zhang, Xiaoyong Du, Jie Song, Peng Li, Fuxin Jiang, Tieying Zhang, Jianjun Chen. DeepPrep: An LLM-Powered Agentic System for Autonomous Data Preparation. VLDB 2026.

DeepPrep：自主数据准备

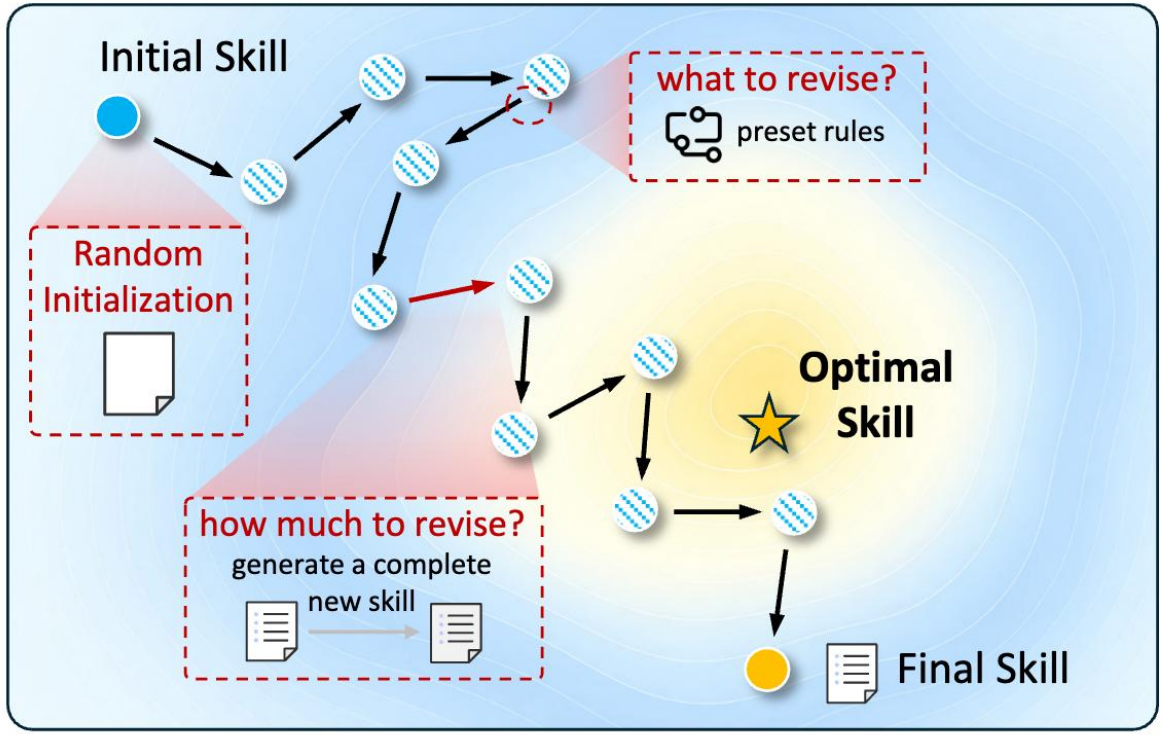
DeepPrep 端到端数据准备效果



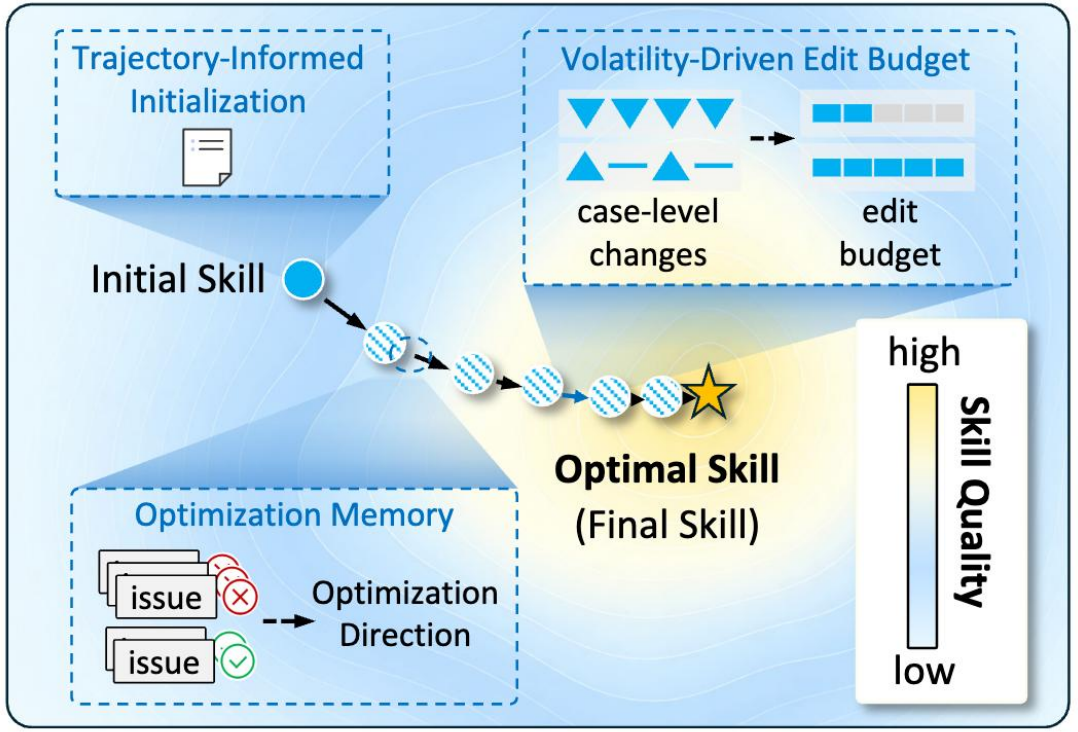
取得成本-质量帕累托最优的数据准备效果

SkillAdam：稳定、高效的Skill自进化

Skill 构建　Skill依赖人工撰写，优化不稳定



(a) Existing SGD-like methods.

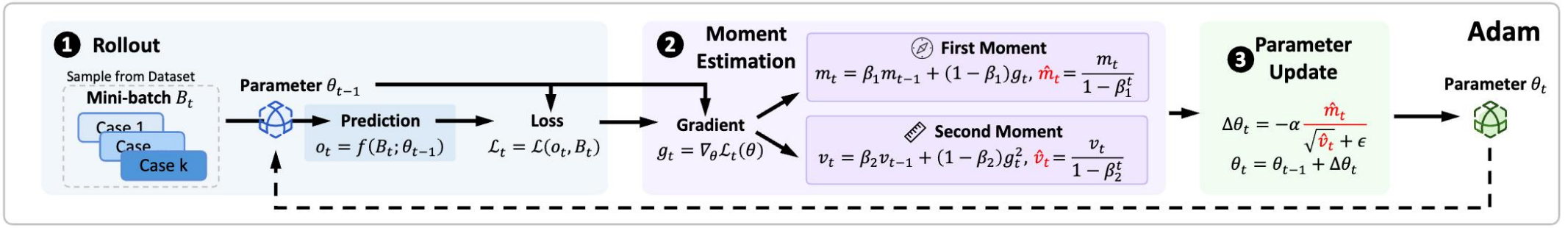


(b) SKILLADAM (ours).

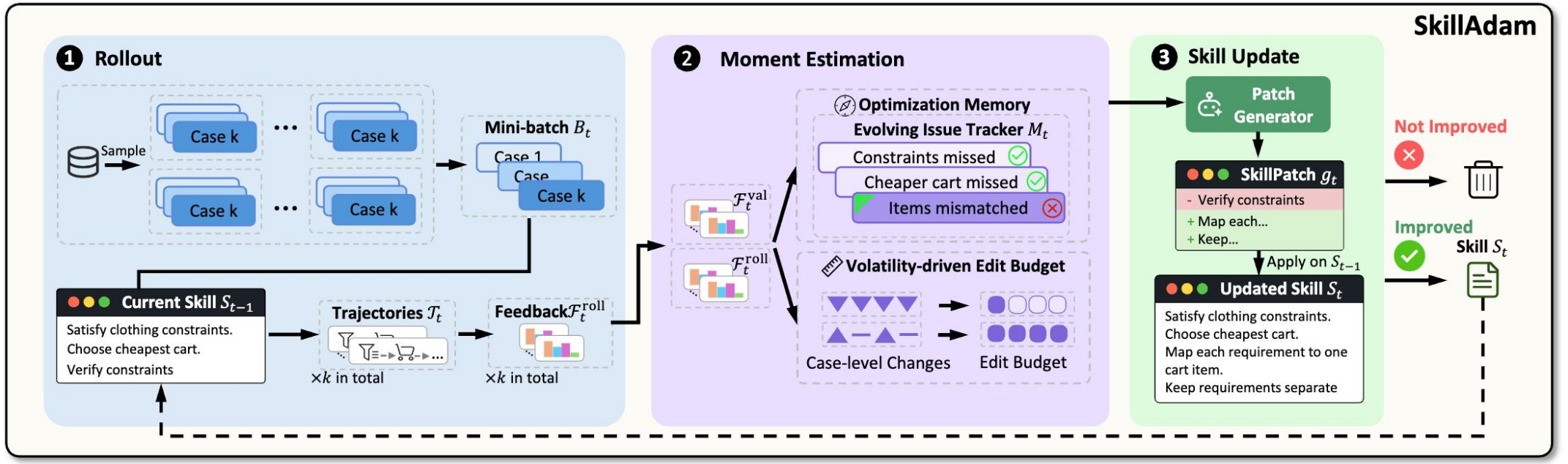
SkillAdam：稳定、高效的Skill自进化

Skill 自进化 从用户trajectory自主沉淀skill

Adam优化模型参数



SkillAdam优化Agent Skill



SkillAdam：稳定、高效的Skill自进化

Skill 自进化 Skill在短程（~10轮）和长程Agent任务上显著提升

短程任务

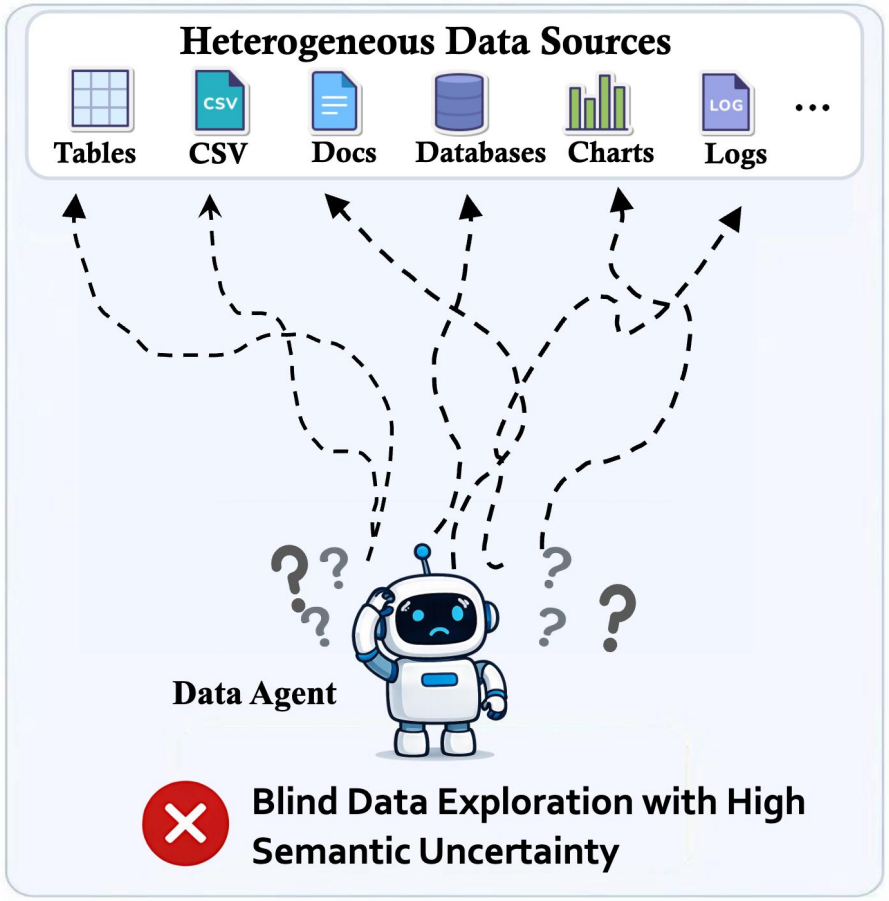
Method	SearchQA	Spreadsheet	OfficeQA	DocVQA	LiveMath
NoSkill	77.7	41.8	33.1	78.8	37.6
HumanSkill	81.8	72.9	66.9	90.1	38.4
LLMSkill	80.9	43.2	51.7	89.6	40.0
Trace2Skill	82.4	49.6	65.7	90.6	52.0
TextGrad	81.4	41.1	42.0	87.2	49.2
GEPA	84.8	73.6	63.9	89.1	43.2
SkillOpt	87.3	80.7	72.1	91.2	66.9
SkillAdam	**87.5**	**81.1**	**72.1**	**92.3**	**67.7**

长程任务

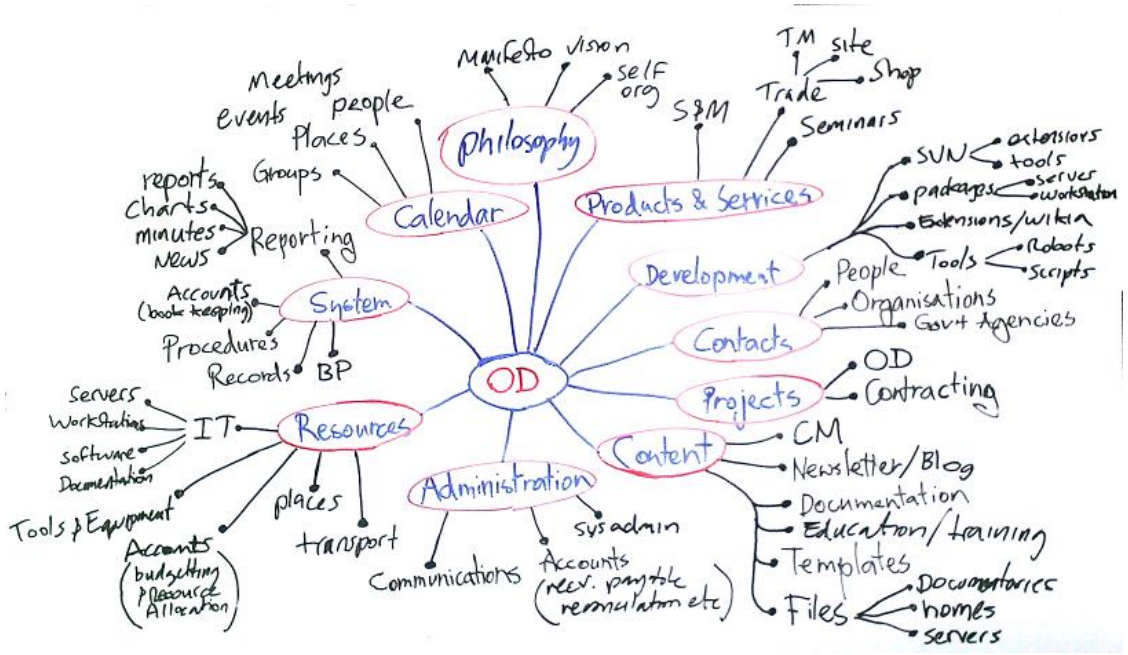
Method	ALFWorld	DP-Shopping	DP-Travel	DP-Avg
NoSkill	83.6	31.7	0.0	15.8
SkillOpt	87.3	41.7	1.7	21.7
SkillAdam	**89.6**	**45.0**	**11.7**	**28.3**

EvoOntology：自进化Ontology层

Agent 在复杂工作区完成任务，需要对环境的语义建模



(a) Data Agent w/o Ontology Layer

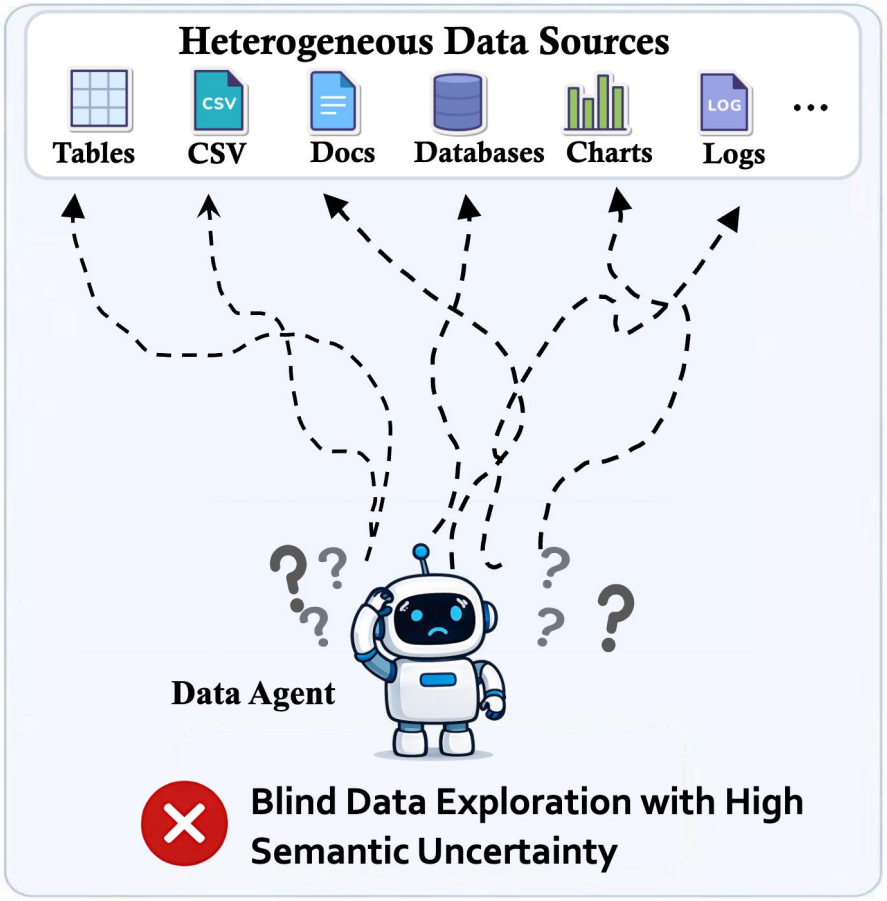


核心问题：探讨“世界上存在什么？” 以及实体、属性、关系、空间和时间之间的基本规律。

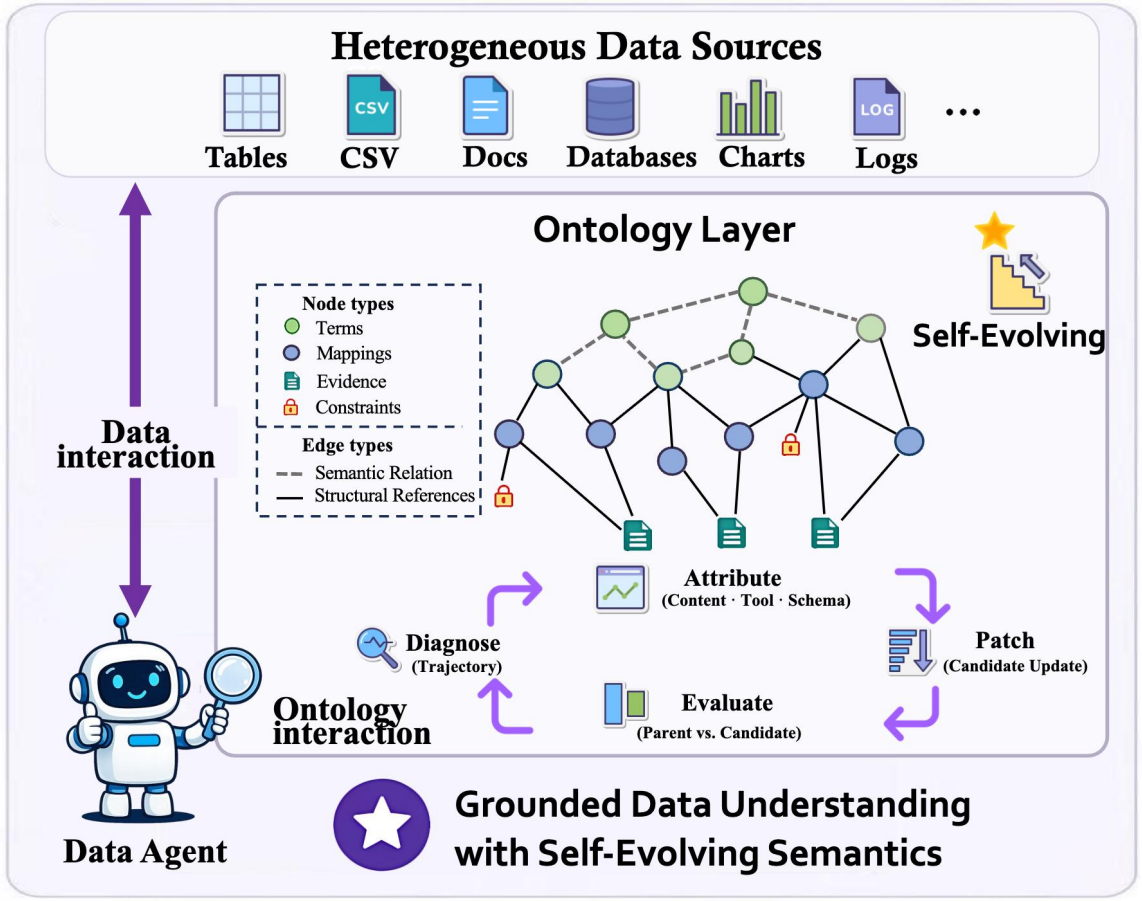
计算机领域基本定义：对特定领域中“概念、数据和实体之间的类别、属性和关系”的**形式化表达与规范化描述**。

EvoOntology：自进化Ontology层

Agent 在复杂工作区完成任务，需要对环境的语义建模

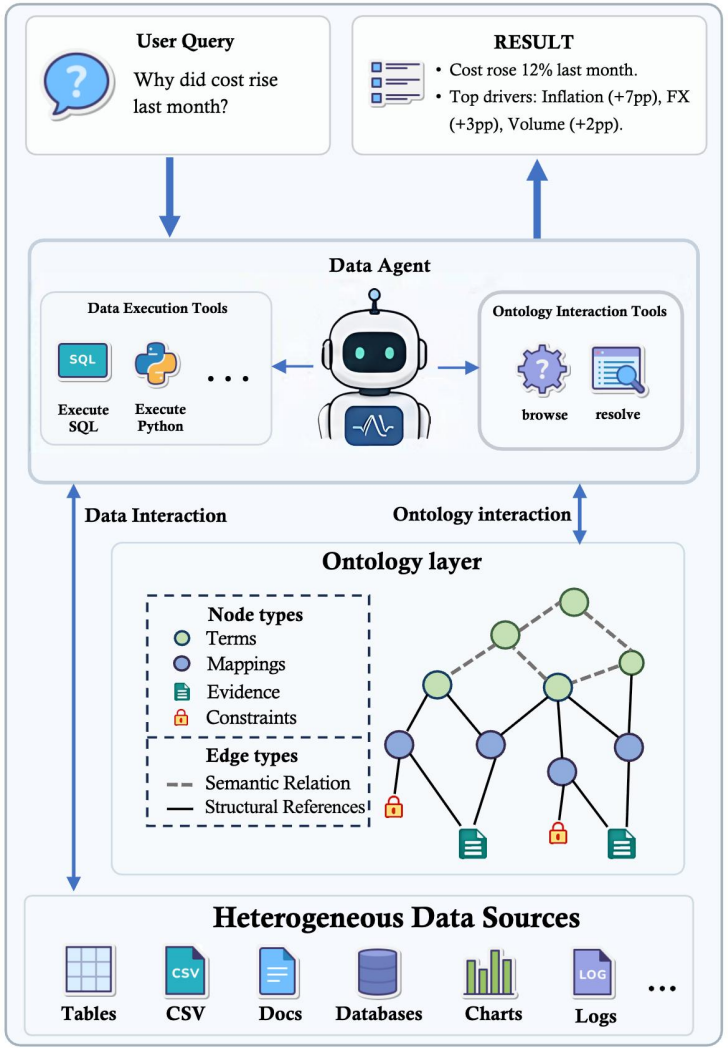


(a) Data Agent w/o Ontology Layer

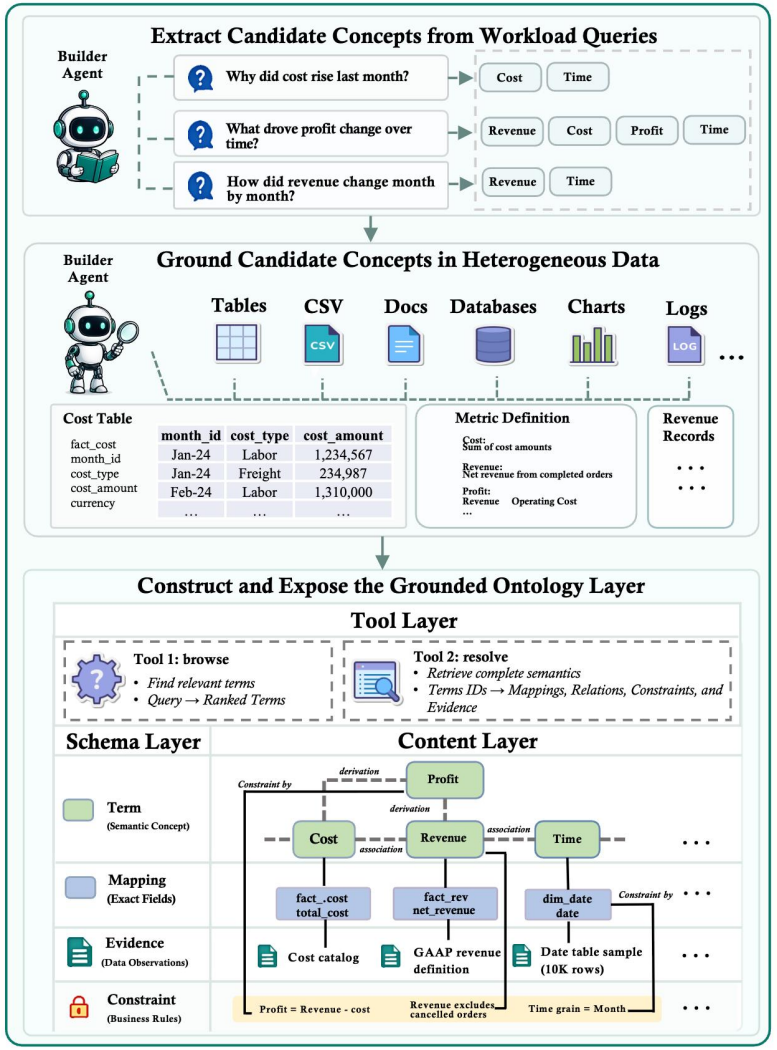


(b) Data Agent with Self-Evolving Ontology Layer

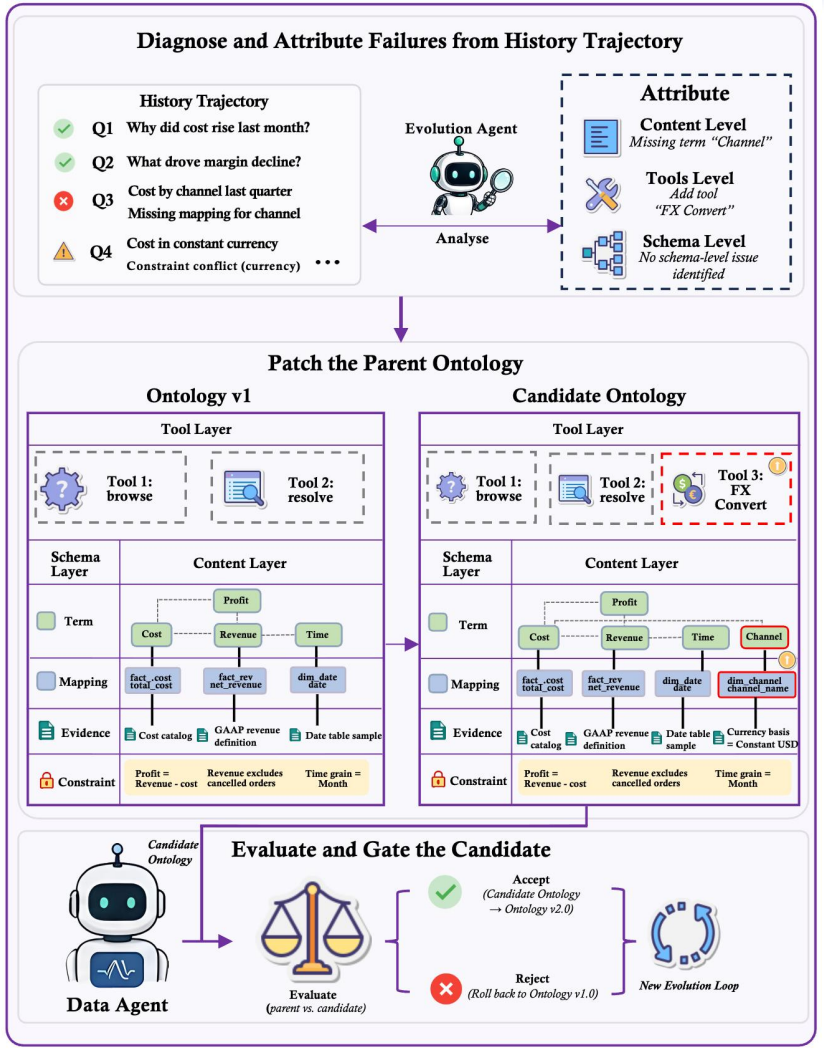
EvoOntology：自进化Ontology层



本体结构



探查环境→初始化本体



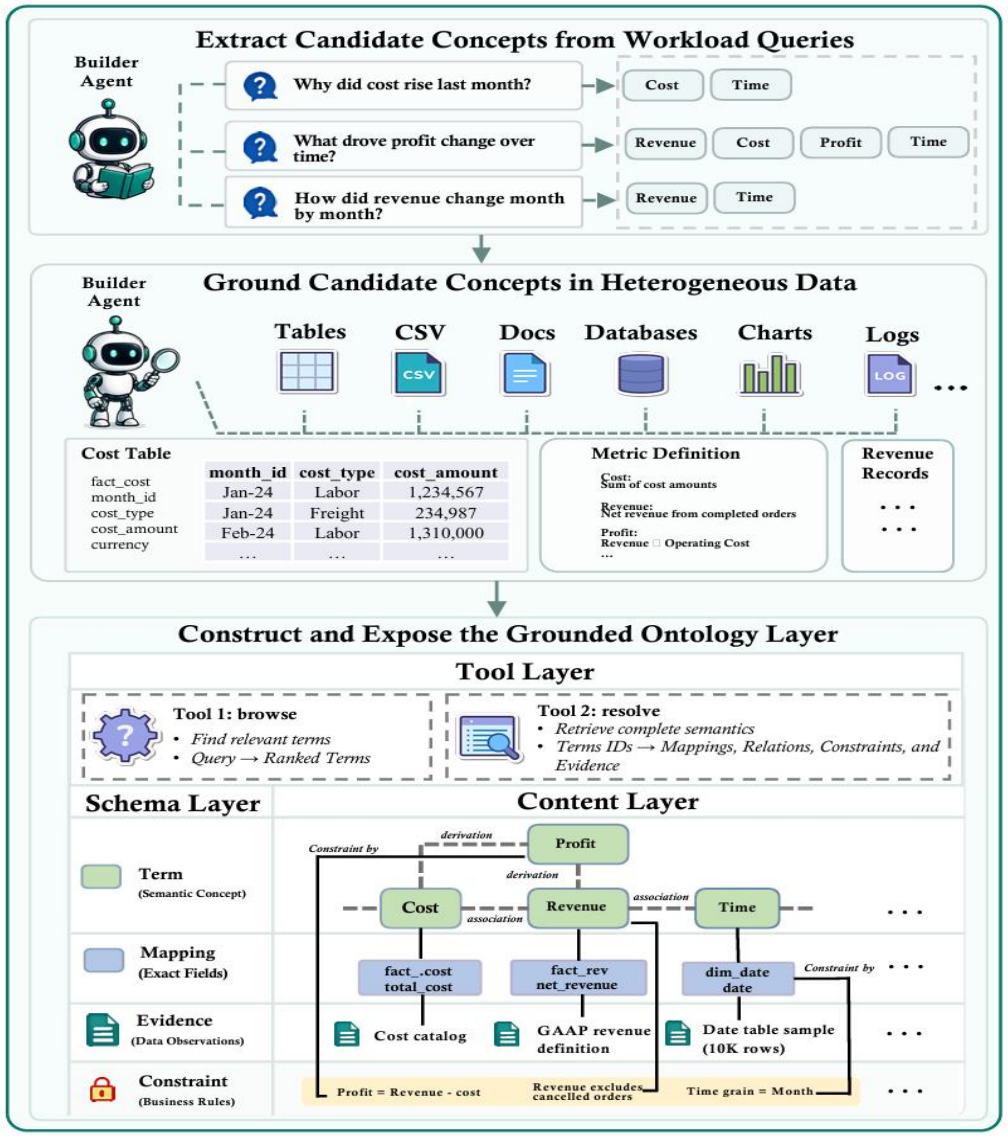
轨迹积累→自进化本体

EvoOntology：自进化Ontology层

构建初始本体

收集任务与数据源，探测数据并保留证据。

层	作用
Content Layer	类型化语义图，包含 Term、Mapping、Constraint 和 Evidence 四类节点。Semantic Relation 连接 Term；Structural Reference 连接 Term 与 Mapping，并将 Constraint 或 Evidence 挂接到其约束或支撑的对象上。
Schema Layer	定义四类节点的字段、允许使用的 Semantic Relation 类型，以及合法的 Structural Reference 模式，从而确定 Ontology Layer 的表达边界。
Tool Layer	通过 browse_semantics、resolve_semantics 和简洁的 session manifest 向 Agent 暴露 Ontology Layer。会话初始化时只注入 manifest；具体记录及其关联对象均按需检索。



EvoOntology：自进化Ontology层

根据反馈进化

收集智能体的执行轨迹，诊断问题并定位修改本体层，验证通过后更新本体层。

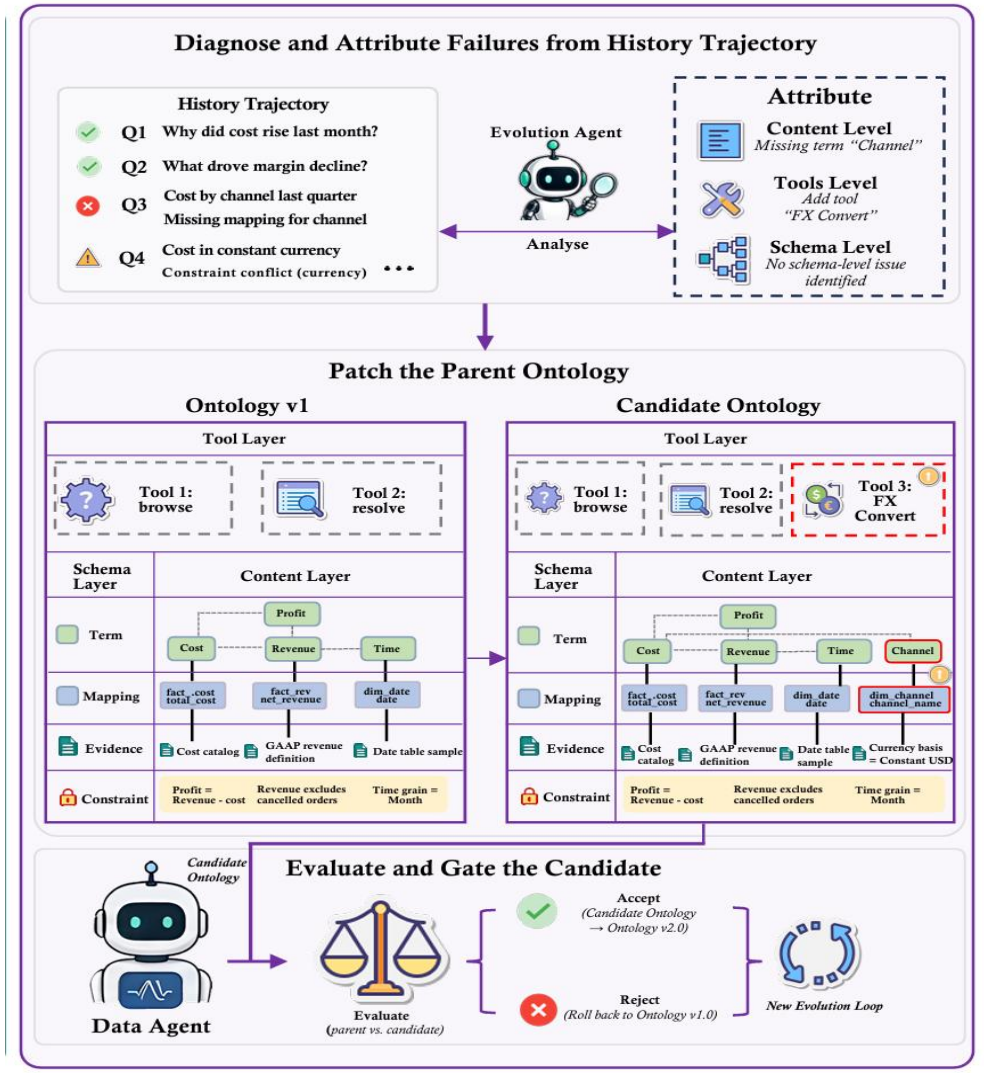
1.Build — 从 workload 提取候选概念，在原始数据源中验证，并发布 ontology_v0。

2.Use — Data Agent 按需查询 Ontology Layer，同时记录工具交互和任务结果。

3.Evolve — 诊断重复出现的行为，将问题归因到 Content、Tool 或 Schema，并生成局部 Candidate 补丁。

4.Evaluate — 在相同数据、Agent、解码配置和交互预算下比较 Parent 与 Candidate。

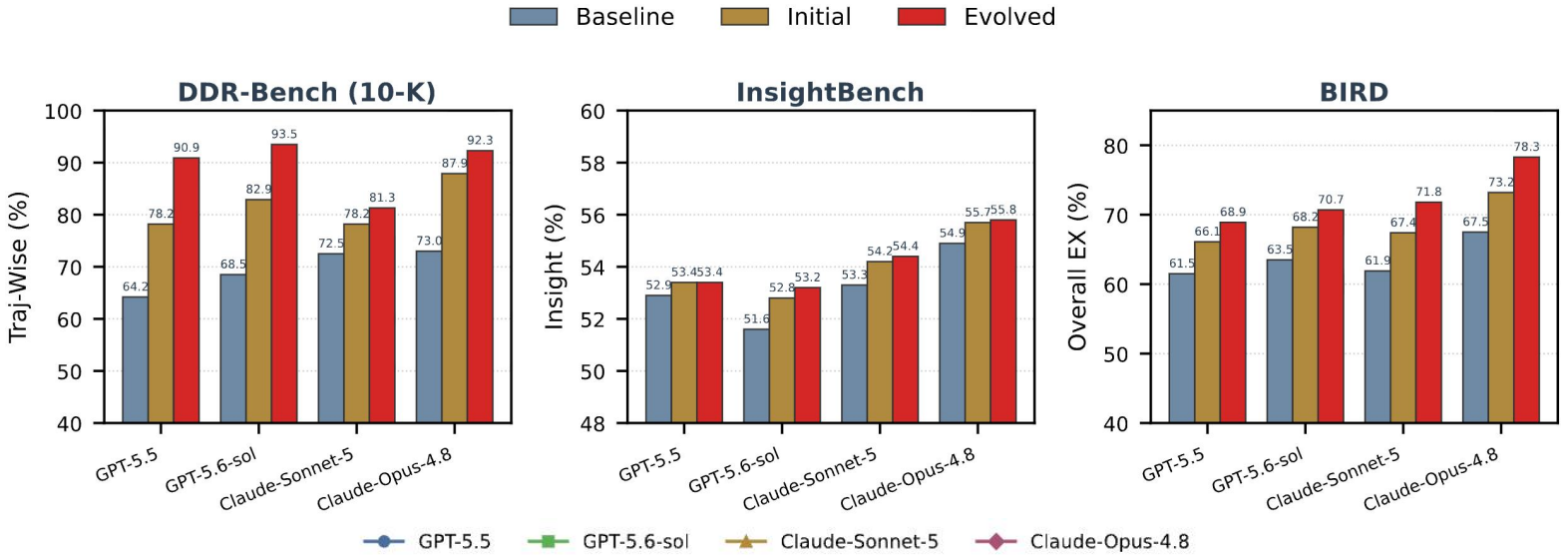
5.Publish or reject — 通过门控的 Candidate 发布为新ontology；否则保留 Parent，并把结果用于下一轮。



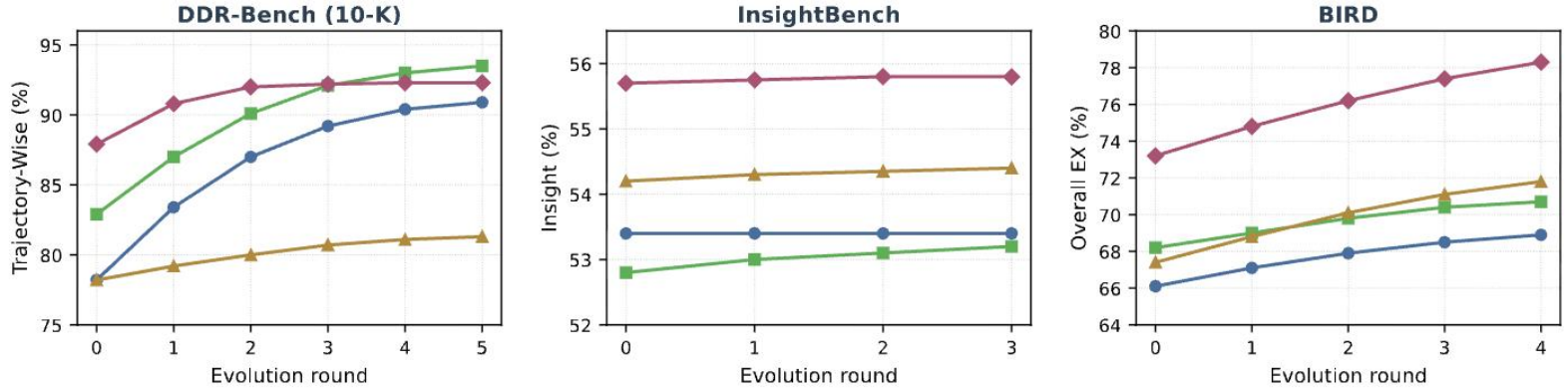
EvoOntology：自进化Ontology层

EvoOntology 自进化Ontology层优于Palantir式本体层

Agent在复杂任务上表现更准确



Agent表现随着Ontology层的进化持续提升



EvoOntology：自进化Ontology层

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总结

Agent-Ontology-Data协同进化

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Python ☆ 1k 89

SkillAdam Public

SkillAdam: Better Skills for Your AI Agent🚀 Claude Code/Codex/Cursor插件，一键进化你的skill

Python ☆ 100 3

CoDA-Bench Public

CoDA-Bench is a benchmark for code agents on data-intensive tasks.
🎈代码智能体能搞定数据密集型任务吗?

Python ☆ 47 1

DataEvolver Public

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谢 谢 大 家

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